

Lab 4: Flights

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Part I

1. What is the unit of observation in the data frame on the handout?

The unit of observation in the data frame of one individual domestic commercial flight in the US.

2. Which variables are categorical?

“flight number”, “carrier”, “origin”, “dest”, “tailnum”, “year”, “hour”, “minute” is categorical

3. Which variables are numerical?

“air_time”, “distance”, “dep_delay”, “arr_delay” is numerical.

4. Do any of the variable have ambiguous data types to you?

“sched_dep_time”, “arr_time”, “sched_arr_time” and “time_hour” are a bit ambiguous, they should be in their own “day-time” category.

5. Is there any discernible pattern to the manner in which the rows are ordered?

The row seems to be ordered by the time of departure, from the earliest day, month, year to the least.

6. What is your guess for the units/format used to record the departure time? Said another way, what would a value of 1517 represent?

It is probably recorded in military time of a 24 hour frame. 1517 would mean 3:17pm.

7. What filter would you use to extract the flights that left in the springtime?

I would filter the planes departure time on or after 000 (12:00am) March 20, 2020 and before on or at 2359 (11:59pm) June 20, 2020.

Part II

1. filter(): Filter the data set to contain only the flights that went to Portland, Oregon and print the first few rows of the data frame. How many were there in 2020?

```
Portland <- filter(flights,(dest == "PDX")&(year == "2020"))
Portland
```

```
## # A tibble: 3,882 x 19
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <dbl> <dbl> <dbl>   <dbl>         <dbl>         <dbl>   <dbl>         <dbl>
## 1  2020     1     1     613             600           13     801             754
```

```
## 2 2020 1 1 656 700 -4 842 854
## 3 2020 1 1 657 700 -3 836 845
## 4 2020 1 1 825 830 -5 1024 1024
## 5 2020 1 1 900 900 0 1116 1050
## 6 2020 1 1 1055 1055 0 1240 1240
## 7 2020 1 1 1110 1115 -5 1327 1319
## 8 2020 1 1 1129 1135 -6 1354 1329
## 9 2020 1 1 1246 1248 -2 1443 1443
## 10 2020 1 1 1304 1305 -1 1518 1459
## # ... with 3,872 more rows, and 11 more variables: arr_delay <dbl>,
## #   carrier <chr>, flight <dbl>, tailnum <chr>, origin <chr>, dest <chr>,
## #   air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dtm>
```

3882 of the observations are in 2020 and is landing at Portland, Oregon.

2. Mutate a new variable called `avg_speed` that is the average speed of the plane during the flight, measured in miles per hour. (Look through the column names or the help file to find variables that can be used to calculate this.)

```
flights <- mutate(flights, avg_speed = distance/(air_time/60))
select (flights, avg_speed)
```

```
## # A tibble: 120,605 x 1
##   avg_speed
##   <dbl>
## 1      516.
## 2      464.
## 3      540.
## 4      524.
## 5      558.
## 6      518.
## 7      519.
## 8      501.
## 9      446.
## 10     560
## # ... with 120,595 more rows
```

3. `arrange()`: Arrange the data set to figure out: which flight holds the record for longest departure delay (in hrs) and what was its destination? What was the destination and delay time (in hrs) for the flight that was least delayed, i.e. that left the most ahead of schedule?

```
flights_dep_delay <- arrange(flights, dep_delay)
view(flights_dep_delay)
```

Flight N191UW from SFO to PHX holds the record for longest departure delay; It delayed by 29 hours. Flight N175SY from SFO to GEG holds the record that was least delayed; it went earlier by 0.66 hour.

4. `summarize()`: Confirm the records for departure delay from the question above by summarizing that variable by its maximum and its minimum value.

```
summarize(flights, max = max(dep_delay, na.rm = TRUE),
          min = min (dep_delay, na.rm = TRUE))
```

```
## # A tibble: 1 x 2
##   max    min
```

```
##    <dbl> <dbl>
## 1  1740   -40
```

It is true, the max and min are shown in minutes in the table above. By dividing the value by 60, we can find the same answer in hours.

5. How many flights left SFO during March 2020?

```
flights %>%
  filter(month == 3, origin == "SFO", year == 2020) %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1 14165
```

14165 flights left SFO during March 2020.

6. How many flights left SFO during April 2020?

```
flights %>%
  filter(month == 4, origin == "SFO", year == 2020) %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1  4517
```

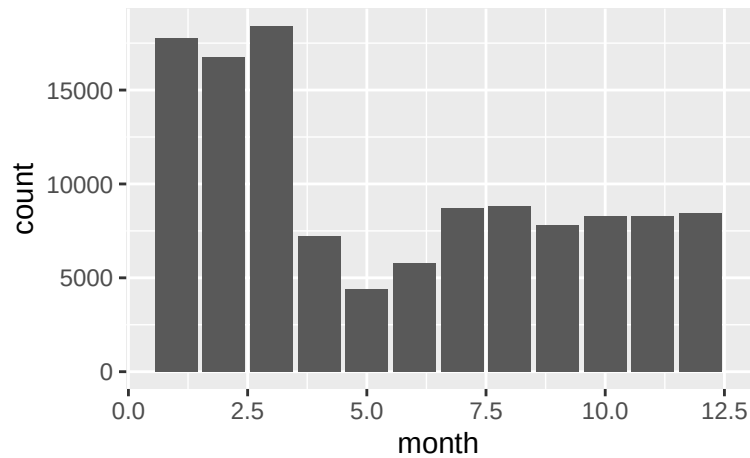
4517 flights left SFO during April 2020.

7. Create a bar chart that shows the distribution by month of all the flights leaving the Bay Area (SFO and OAK). Do you see any sign of an effect of the pandemic?

```
flights %>%
  filter((origin == "SFO" | origin == "OAK")) %>%
  group_by(month) %>%
  summarise(n = n())
```

```
## # A tibble: 12 x 2
##   month     n
##   <dbl> <int>
## 1     1 17780
## 2     2 16748
## 3     3 18415
## 4     4  7224
## 5     5  4399
## 6     6  5756
## 7     7  8697
## 8     8  8815
## 9     9  7786
## 10    10  8257
## 11    11  8296
## 12    12  8432
```

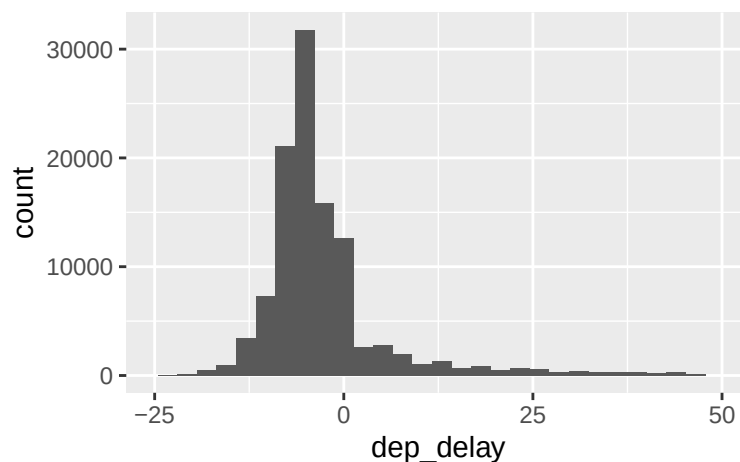
```
ggplot(flights, aes(x = month)) +  
  geom_bar()
```



Yes, the graph is right skewed to an extent, meaning that the pandemic have affected the flights . March (3.0) was when the lock down first being implemented in the United State, a dramatic drop in the number of flights following March can be seen. May (5.0) has the lowest count in flights departing from San Francisco Although slowly recovered following June to December (6.0 - 12.0), still not able to achieve the same number of flights as before pandemic (January to March 1.0 -3.0).

8. Create a histogram showing the distribution of departure delays for all flights. Describe in words the shape and modality of the distribution and, using numerical summaries, (i.e. summary statistics) its center and spread. Be sure to use measures of center and spread that are most appropriate for this type of distribution. Also set the limits of the x-axis to focus on where most of the data lie.

```
ggplot(flights, aes(x = dep_delay)) + geom_histogram() + xlim(-25,50)
```



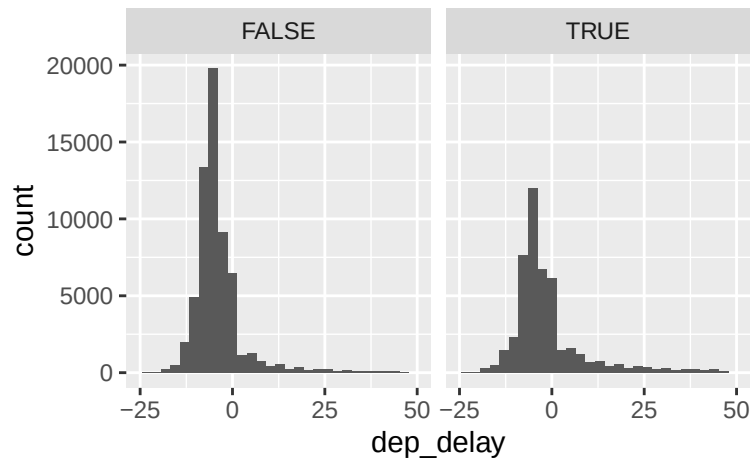
```
summarise(flights,  
  med = median(dep_delay, na.rm = TRUE),  
  iqr = IQR(dep_delay, na.rm = TRUE))
```

```
## # A tibble: 1 x 2  
##   med   iqr  
##   <dbl> <dbl>  
## 1    -4     6
```

The graph seems to be right skewed and unimodal distributed. The center is located at -4 minutes (the median) and the interquartile range is 6, meaning that the center 50% of the planes have departure delay ranging from -7 to -1 minutes. Most departure delay seems to be under 0, meaning that most planes left on time or even before the scheduled time. There's fewer planes that have positive departure delay comparing to negative departure delay. As the minutes of departure delay increases, the count of planes decrease (fluctuated in between).

9. Add a new column to your data frame called `before_times` that takes values of `TRUE` and `FALSE` indicating whether the flight took place up through the end of March or after April 1st, respectively. Remake the histograms above, but now separated into two subplots: one with the departure delays from the before times, the other with the flights from afterwards. Can you visually detect any difference in the distribution of departure delays?

```
flights %>%
  mutate (before_times = month<4) %>%
  ggplot(aes(dep_delay)) + geom_histogram() + facet_wrap(vars(before_times)) + xlim(-25,50)
```



Both graphs seem to have a right skewed and unimodal distribution. However, the center position is different. From April first and on wards, the departure delay were shorter (center moves to left), at about -6.25. Whereas for before April, the center of departure delay is about -2. In addition, the “True” graph has less volume (counts) than the “False” graph at the < 0 range; and the “True” graph has more volume for positive departure delay (0 - 50 minutes) than the “False” graph.

10. If you flew out of OAK or SFO during this time period, what is the tail number of the plane that you were on? If you did not fly in this period, find the tail number of the plane that flew JetBlue flight 40 to New York’s JFK Airport from SFO on May 1st.

```
select(filter(flights, month == 5, day == 1, flight == 40, dest == "JFK", origin == "SFO"), tailnum)

## # A tibble: 1 x 1
##   tailnum
##   <chr>
## 1 N982JB

It is N982JB.
```

11. What proportion of the flights left on or ahead of schedule?

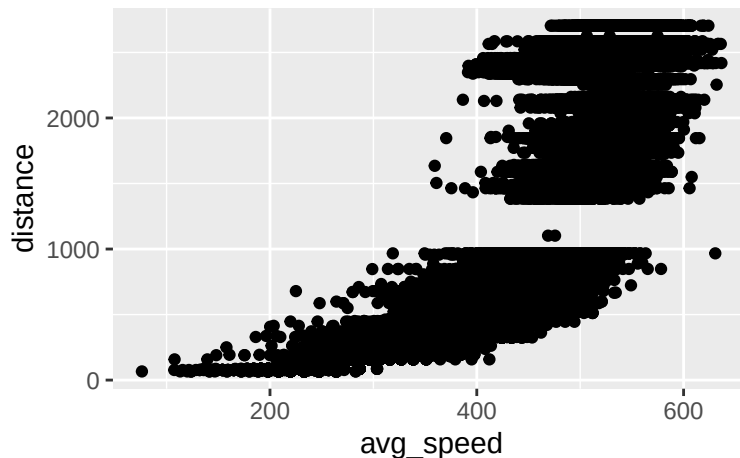
```
flights %>%
  filter(dep_delay <= 0) %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1 91873
```

91873 flights left on or ahead of schedule, which is about 76.18% of the planes.

12. Create a plot that captures the relationship of average speed vs. distance and describe the shape and structure that you see. What phenomena related to taking flights from the Bay Area might explain this structure?

```
flights <- mutate(flights, avg_speed = distance/(air_time/60))
ggplot(flights, aes (x= avg_speed, y = distance)) + geom_point()
```



It looks like there's a jump between the 1000 - 1400 distance range. This could be due to the fact that some of the flights have to cross the rocky mountain, so the most distance is either above 1400 miles or below 1000 miles. I also see the trend that the average speed increases as the distance increases - distance 0 - 1000 miles has a average speed range around 0 -630 mile/hour; distance from 1400 - 2400 have an average speed range around 380 -630 mile/hour.

13. For OAK and SFO separately, what proportion of the flights left on or ahead of schedule?

```
flights %>%
  filter(origin == "SFO") %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1 89394
```

```
flights %>%
  filter(dep_delay <= 0, origin == "SFO") %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
```

```
## <int>
## 1 69127

flights %>%
  filter(origin == "OAK") %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1 31211
```

```
flights %>%
  filter(dep_delay <= 0, origin == "OAK") %>%
  summarise(count = n())
```

```
## # A tibble: 1 x 1
##   count
##   <int>
## 1 22746
```

There are 31211 planes departed from OAK total, and 22746 left on or ahead of schedule, so 72.94% left on or ahead of schedule on OAK.

There are 89394 planes departed from SFO total, and 69127 left on or ahead of schedule, so 77.33% left on or ahead of schedule on SFO.

14. Create a data frame that contains the median and interquartile range for departure delays, grouped by carrier. Which carrier has the lowest typical departure delay? Which one has the least variable departure delays?

```
flights %>%
  group_by(carrier) %>%
  summarise(
    med = median(dep_delay, na.rm = TRUE),
    iqr = IQR(dep_delay, na.rm = TRUE)) %>%
  arrange(med)
```

```
## # A tibble: 12 x 3
##   carrier med iqr
##   <chr>   <dbl> <dbl>
## 1 B6      -8     7
## 2 AS      -7     9
## 3 G4      -7    13.8
## 4 YV      -7    11
## 5 AA      -6     6
## 6 DL      -5     5
## 7 F9      -5     9
## 8 HA      -5     9
## 9 OO      -5     7
## 10 UA     -5     6
## 11 NK     -4     6
## 12 WN     -3     5
```

```
flights %>%
  group_by(carrier) %>%
  summarise(
    med = median(dep_delay, na.rm = TRUE),
    iqr = IQR(dep_delay, na.rm = TRUE)) %>%
```

```
arrange(iqr)
```

```
## # A tibble: 12 x 3
##   carrier med iqr
##   <chr>   <dbl> <dbl>
## 1 DL      -5     5
## 2 WN      -3     5
## 3 AA      -6     6
## 4 NK      -4     6
## 5 UA      -5     6
## 6 B6      -8     7
## 7 OO      -5     7
## 8 AS      -7     9
## 9 F9      -5     9
## 10 HA     -5     9
## 11 YV     -7    11
## 12 G4     -7   13.8
```

B6 carrier has the lowest typical departure, and DL has the least variable departure delay.

PART III

1. For flights leaving SFO, which month has the highest average departure delay? What about the highest median departure delay? Which of these measures is more reliable for deciding which month(s) to avoid flying if you particularly dislike flights that are severely delayed.

```
flights %>%
  filter (origin == "SFO") %>%
  group_by (month) %>%
  summarise (avg_delay = mean(dep_delay, na.rm = TRUE)) %>%
  arrange(avg_delay)
```

```
## # A tibble: 12 x 2
##   month avg_delay
##   <dbl>   <dbl>
## 1     4    -3.81
## 2     5    -2.68
## 3    11    -1.91
## 4     6    -1.86
## 5     8    -1.80
## 6     9    -1.75
## 7     7    -1.56
## 8    12    -0.970
## 9    10    -0.155
## 10     3     1.15
## 11     2     8.06
## 12     1     9.05
```

```
flights %>%
  filter (origin == "SFO") %>%
  group_by (month) %>%
  summarise (median_delay = median (dep_delay, na.rm = TRUE)) %>%
  arrange(median_delay)
```

```
## # A tibble: 12 x 2
```

```
##      month median_delay
##      <dbl>         <dbl>
## 1         4           -7
## 2         5           -6
## 3         7           -6
## 4        11           -6
## 5         3           -5
## 6         6           -5
## 7         8           -5
## 8         9           -5
## 9        10           -5
## 10        12           -5
## 11         1           -3
## 12         2           -3
```

March has the highest average departure delay of 1.15 minutes. December has the highest median departure delay of -5 minutes. Mean is a better choice for someone who wants to avoid flight delay, because mean is more sensitive to the extreme values. In addition, the mean is larger than the median, meaning that most planes left on time or even before the schedule time, there are some extreme values that result the mean to be positive and larger than the median (negative value).

2. Each individual airplane can be uniquely identified by its tailnumber in the same way that US citizens can be by their social security numbers. Which airplane flew the farthest during this year for which we have data? How many times around the planet does that translate to?

```
arrange (flights, desc(distance))

## # A tibble: 120,605 x 20
##   year month   day dep_time sched_dep_time dep_delay arr_time sched_arr_time
##   <dbl> <dbl> <dbl>   <dbl>         <dbl>         <dbl>   <dbl>         <dbl>
## 1  2020     1     1     718             720           -2     1558             1600
## 2  2020     1     1     812             820           -8     1629             1652
## 3  2020     1     1     830             830            0     1712             1703
## 4  2020     1     1    1042            1040            2     1910             1915
## 5  2020     1     1    1410            1414           -4     2230             2247
## 6  2020     1     1    1436            1440           -4     2257             2309
## 7  2020     1     1    2036            2050          -14     449              517
## 8  2020     1     1    2152            2200           -8     613              625
## 9  2020     1     1    2217            2220           -3     746              705
## 10 2020     1     1    2347            2350           -3     814              819
## # ... with 120,595 more rows, and 12 more variables: arr_delay <dbl>,
## #   carrier <chr>, flight <dbl>, tailnum <chr>, origin <chr>, dest <chr>,
## #   air_time <dbl>, distance <dbl>, hour <dbl>, minute <dbl>, time_hour <dtm>,
## #   avg_speed <dbl>
```

There seems to have multiple tailnum with the same farthest distance at 2704 miles. Earth's circumference is 24,901, so that will be 0.109 times around the planet.

3. What is the tailnumber of the fastest plane in the dataset? What type of plane is it (google it!)? Be sure to be clear how you're defining fastest.

```
flights <- mutate(flights, avg_speed = distance/(air_time/60))
arrange(flights, desc(avg_speed)) %>%
select (tailnum, avg_speed, origin, dest)
```

```
## # A tibble: 120,605 x 4
##   tailnum avg_speed origin dest
##   <chr>      <dbl> <chr>  <chr>
## 1 N30913      637. SFO    IAD
## 2 N213UA      636. SFO    EWR
## 3 N2142U      633. SFO    EWR
## 4 N17002      633. SFO    EWR
## 5 N881UA      632. SFO    PIT
## 6 N788UA      631. SFO    IAD
## 7 N354FR      631. SFO    DEN
## 8 N77431      628. SFO    IAD
## 9 N27908      628. SFO    IAD
## 10 N209UA      628. SFO    IAD
## # ... with 120,595 more rows
```

N30913 is the fastest plane in the data set with an average speed of 636.5789 miles/hour. It is a Boeing 787-8 Dreamliner! I choose average speed because average is more sensitive to extreme speed such as during acceleration and deceleration of the flight.

4. Using the airport nearest your hometown, which day of the week and which airline seems best for flying there from San Francisco?

```
library(lubridate)
flights %>%
  filter(origin == "SFO" & dest == "LAX") %>%
  mutate(day_of_week = wday(ymd(paste(year, month, day, set = "-")), label = T)) %>%
  group_by(day_of_week) %>%
  summarise(avg_delay = mean(arr_delay, na.rm = TRUE), med_delay = median(arr_delay, na.rm = TRUE),
            arrange(desc(avg_delay)))
```

```
## # A tibble: 7 x 4
##   day_of_week avg_delay med_delay iqr_delay
##   <ord>      <dbl>      <dbl>      <dbl>
## 1 Fri        -11.8        -19         20
## 2 Thu        -12.6        -20         20
## 3 Sun        -12.6        -20         19
## 4 Sat        -13.6        -21         19
## 5 Wed        -16.1        -21         17
## 6 Mon        -16.2        -21         19
## 7 Tue        -18.3        -23         17
```

```
library(lubridate)
flights %>%
  filter(origin == "SFO" & dest == "LAX") %>%
  mutate(day_of_week = wday(ymd(paste(year, month, day, set = "-")), label = T)) %>%
  group_by(carrier) %>%
  summarise(avg_delay = mean(arr_delay, na.rm = TRUE), med_delay = median(arr_delay, na.rm = TRUE),
            arrange(desc(avg_delay)))
```

```
## # A tibble: 7 x 4
##   carrier avg_delay med_delay iqr_delay
##   <chr>      <dbl>      <dbl>      <dbl>
## 1 AS         0.458         -9         26
## 2 AA        -0.867        -14         16
## 3 DL        -9.21        -15         17
```

## 4 UA	-16.3	-22	16
## 5 00	-17.2	-22	17
## 6 B6	-19	-24	14.8
## 7 WN	-21.6	-26	14

It seems to be best to fly from SFO to LAX on Tuesday, because Tuesday have the lowest average arrival delay of -18.26 minute, meaning that I'll land 18.36 minutes earlier than the expected flight time on average. Its median of arrival delay is also the lowest at -23 minutes. The median is smaller than the mean, which shows that the distribution of the data is right skewed and that most data are around -23 minutes. Moreover, its IQR of 17 minutes are among the lowest, meaning that on Tuesday the arrival delay are the least spread from the mean arrival delay: the chance of extreme delays on Tuesday are lower compared to other days. I choose this because the "best" way for me to fly is to get home from SFO as soon as possible and with great certainty.

And the best airline to fly with is WN, because it has the lowest arrival delay of -21.58 minutes and lowest iqr delay of 14 minutes. This shows that WN provides the quickest time to be arrived and less susceptible for extreme delays.

5. Provide a code that will show the graph below.

```
flights %>%
  filter (origin == "SFO") %>%
  group_by(day, month) %>%
  mutate(avg_delay= mean (dep_delay, na.rm = TRUE), count = n()) %>%
ggplot(aes(x = count, y = avg_delay),
        color = month) +
  geom_point(alpha= 0.7) +
  labs(title = "Days with more flights have more delays",
        color = "species")
```

